

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Classified Practice

Arithmetic Sequences

Q001 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q005 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q007 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-2, 1, 5)$

B $(-2, 1, 4)$

C $(-2, 0, 4)$

D $(2, -1, -4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (11, 15, 20)

B (11, 14, 19)

C (11, 15, 19)

D (-11, -15, -19)

Classified Practice

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 2, 4)

B (3, 1, 3)

C (-3, -2, -3)

D (3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (5, 11, 21)

B (5, 11, 22)

C (5, 10, 21)

D (-5, -11, -21)

Classified Practice

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 10, 17)

B (4, 10, 16)

C (4, 9, 16)

D (-4, -10, -16)

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 1, 0)

B (3, 0, -1)

C (3, 1, -1)

D (-3, -1, 1)

Classified Practice

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (4, 3, 7)

B (4, 2, 6)

C (-4, -3, -6)

D (4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, -3, -1)$

B $(-3, -3, 0)$

C $(-3, -4, -1)$

D $(3, 3, 1)$

Classified Practice

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (1, 10, 26)

B (1, 10, 25)

C (1, 9, 25)

D (-1, -10, -25)

Q018 MCQ**For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** (3, 6, 12)**B** (3, 5, 11)**C** (3, 6, 11)**D** (-3, -6, -11)

Classified Practice

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-3, 4, 12)$

B $(-3, 3, 11)$

C $(3, -4, -11)$

D $(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 9, 20)

B (2, 9, 21)

C (2, 8, 20)

D (-2, -9, -20)

Classified Practice

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 4, 8)

B (2, 4, 9)

C (2, 3, 8)

D (-2, -4, -8)

Q026 MCQ**For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-2, 4, -7)$ **B** $(-2, 4, -8)$ **C** $(-2, 3, -8)$ **D** $(2, -4, 8)$

Classified Practice

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (6, 18, 55)

B (6, 17, 54)

C (6, 18, 54)

D (-6, -18, -54)

Q028 MCQ**For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:****Choose the correct option.****A** $(-15, 45, -134)$ **B** $(-15, 44, -135)$ **C** $(15, -45, 135)$ **D** $(-15, 45, -135)$

Classified Practice

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (3, 15, 63)

B (3, 15, 64)

C (3, 14, 63)

D (-3, -15, -63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A (2, 6, 19)

B (2, 6, 18)

C (2, 5, 18)

D (-2, -6, -18)

Classified Practice

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(5, -25, 126)$

B $(5, -26, 125)$

C $(5, -25, 125)$

D $(-5, 25, -125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(-1, 11, -24)$

B $(-1, 10, -25)$

C $(1, -11, 25)$

D $(-1, 11, -25)$

Classified Practice

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Classified Practice

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{2}{3}, \frac{3}{4}, \frac{4}{6})$

B $(\frac{2}{3}, \frac{3}{3}, \frac{4}{5})$

C $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5})$

D $(-\frac{2}{3}, -\frac{3}{4}, -\frac{4}{5})$

Classified Practice

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(\frac{1}{4}, \frac{3}{7}, \frac{1}{3})$

B $(\frac{1}{4}, \frac{3}{6}, \frac{1}{2})$

C $(-\frac{1}{4}, -\frac{3}{7}, -\frac{1}{2})$

D $(\frac{1}{4}, \frac{3}{7}, \frac{1}{2})$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{5}{3}, \frac{5}{2})$

B $(1, \frac{5}{3}, \frac{5}{3})$

C $(1, \frac{5}{2}, \frac{5}{2})$

D $(-1, -\frac{5}{3}, -\frac{5}{2})$

Classified Practice

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(3, 2, \frac{9}{6})$

B $(3, 2, \frac{9}{5})$

C $(3, 1, \frac{9}{5})$

D $(-3, -2, -\frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{7}{6}, \frac{9}{8})$

B $(1, \frac{7}{5}, \frac{9}{7})$

C $(1, \frac{7}{6}, \frac{9}{7})$

D $(-1, -\frac{7}{6}, -\frac{9}{7})$

Classified Practice

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Choose the correct option.

A $(1, \frac{1}{4}, \frac{3}{8})$

B $(1, \frac{1}{3}, \frac{3}{7})$

C $(-1, -\frac{1}{4}, -\frac{3}{7})$

D $(1, \frac{1}{4}, \frac{3}{7})$

Q044 Written

For the sequence $\{a_n\}$: 0, 1, 4, 9, 16, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q045 **Written**

For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.

Q046 Written

For the sequence $\{a_n\}$: 1, 4, 9, 16, 25, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q047 Written

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Q048 MCQ

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Choose the correct option.

A n^2

B n^3

C n^5

D n^6

Classified Practice

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:

Choose the correct option.

A $(n + 2)^3$

B $(n + 2)^4$

C $(n + 2)^2$

D $(n + 2)^5$

Q050 MCQ

Which formula gives the general term of $\{a_n\} : -1, 0, 1, 8, \dots$?:

Choose the correct option.

A $(n - 2)^2$

B $(n - 2)^5$

C $(n - 2)^6$

D $(n - 2)^3$

Classified Practice

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:

Choose the correct option.

A $(n + 5)^2$

B $(n + 5)^3$

C $(n + 5)^4$

D $(n + 5)^5$

Q052 MCQ

Which formula gives the general term of $\{a_n\} : 1, 9, 25, 49, \dots$?:

Choose the correct option.

A $(2n + 1)^2$

B $(2n - 1)^2$

C $(2n - 1)^4$

D $(2n - 1)^5$

Classified Practice

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:

Choose the correct option.

A $(n + 1)^4$

B $(n - 1)^6$

C $(n - 1)^4$

D $(n - 1)^7$

Q054 Written

For the sequence $\{a_n\}$: 2, 5, 8, 11, 14, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q055 **Written**

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q057 Written

For the sequence $\{a_n\} : 4, 9, 14, 19, 24, \dots$, deduce the general term.

Q058 Written

For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.

Classified Practice

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:

Choose the correct option.

A $5n + 2$

B $(5n + 2) + 1$

C $(5n + 2) - 1$

D $-(5n + 2)$

Q060 MCQ**Which formula gives the general term of $\{a_n\} : -4, -1, 2, 5, \dots$?:****Choose the correct option.**

A $(3n - 7) + 1$

B $3n - 7$

C $(3n - 7) - 1$

D $-(3n - 7)$

Classified Practice

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2, ...?:

Choose the correct option.

A $(14 - 4n) + 1$

B $(14 - 4n) - 1$

C $14 - 4n$

D $-(14 - 4n)$

Q062 MCQ

Which formula gives the general term of $\{a_n\} : \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$?:

Choose the correct option.

A $\left(\frac{n}{2}\right) + 1$

B $\left(\frac{n}{2}\right) - 1$

C $-\left(\frac{n}{2}\right)$

D $\frac{n}{2}$

Classified Practice

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Q064 Written

For the sequence $\{a_n\}$: $-3, 6, -9, 12, -15, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q065 Written

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

Q066 Written

For the sequence $\{a_n\}$: $-1, 4, -7, 10, -13, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q067 **Written**

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Classified Practice

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\}$: 1×3 , 2×9 , 3×27 , 4×81 , 5×243 , ..., deduce the general term.

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Classified Practice

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Q074 Written**Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,**

Classified Practice

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (4, 4)
- B (4, 5)
- C (5, 4)
- D (4, -5)

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A $(-7, 4)$
- B $(-7, -4)$
- C $(7, 4)$
- D $(-3, 4)$

Classified Practice

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

- A (18, 5)
 - B (13, -5)
 - C (18, -5)
 - D (-18, -5)
-
-
-

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(12, 4)$

B $(8, -4)$

C $(-12, -4)$

D $(12, -4)$

Classified Practice

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

Classified Practice

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{1}{2}, \frac{2}{3})$

B $(\frac{1}{2}, \frac{1}{3})$

C $(\frac{1}{3}, \frac{1}{2})$

D $(-\frac{1}{2}, \frac{1}{3})$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{3}{4}, \frac{1}{2})$

B $(-\frac{3}{4}, -\frac{1}{2})$

C $(\frac{3}{4}, \frac{1}{2})$

D $(-\frac{1}{4}, \frac{1}{2})$

Classified Practice

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(\frac{5}{3}, \frac{2}{3})$

B $(2, \frac{1}{3})$

C $(\frac{5}{3}, \frac{1}{3})$

D $(-\frac{5}{3}, \frac{1}{3})$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Choose the correct option.

A $(-\frac{5}{2}, -\frac{3}{2})$

B $(-\frac{5}{2}, \frac{3}{2})$

C $(\frac{5}{2}, \frac{3}{2})$

D $(-1, \frac{3}{2})$

Classified Practice

Arithmetic Sequences

Q084 **Written**

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Classified Practice

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 10, B = 18$

B $d = 8, A = 14, B = 22$

C $d = 4, A = 12, B = 18$

D $d = -4, A = 2, B = 10$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 4, A = 9, B = 13$

B $d = -4, A = 9, B = 1$

C $d = 4, A = 1, B = 13$

D $d = 2, A = 3, B = 11$

Classified Practice

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Choose the correct option.

A $d = 3, A = 14, B = 17$

B $d = 4, A = 15, B = 19$

C $d = 4, A = 14, B = 17$

D $d = -4, A = 7, B = 3$

Classified Practice

Arithmetic Sequences

Q089 **Written**

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Classified Practice

Arithmetic Sequences

Q090 **Written**

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Classified Practice

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Classified Practice

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75 .

Classified Practice

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Classified Practice

Arithmetic Sequences

Q098 **Written**

The first term is -8 , and the 8th term is 55 . Find a_n .

Q099 Written

The first term is -18 , and the 10th term is -63 . Find a_n .

Classified Practice

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(3n - 1, 3)$

B $(4n, 4)$

C $(3n + 1, 4)$

D $(a_n, d) = (3n + 1, 3)$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(4n - 3, 4)$

B $(a_n, d) = (4n - 7, 4)$

C $(5n - 8, 5)$

D $(4n - 7, -4)$

Classified Practice

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(a_n, d) = (16 - 4n, -4)$

B $(12 - 4n, -4)$

C $(16 + 4n, 4)$

D $(16 - 3n, -3)$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Choose the correct option.

A $(5n, 5)$

B $(a_n, d) = (4n + 1, 4)$

C $(4n - 1, 4)$

D $(4n + 1, -4)$

Classified Practice

Arithmetic Sequences

Q104 **Written**

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Classified Practice

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Classified Practice

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (9 - 2n, 7)$

B $(7 - 2n, 9)$

C $(1 - 2n, 3)$

D $(9 + 2n, 7)$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n - 3, -2)$

B $(17 + 5n, 2)$

C $(13 - 5n, 8)$

D $(a_n, a) = (5n - 13, -8)$

Classified Practice

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(4n + 2, -2)$

B $(2n + 4, 6)$

C $(a_n, a) = (4n - 2, 2)$

D $(4n - 2, 4)$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(3 - 3n, 6)$

B $(6 + 3n, 3)$

C $(6 - 2n, 2)$

D $(a_n, a) = (6 - 3n, 3)$

Classified Practice

Arithmetic Sequences

Q112 **Written**

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Classified Practice

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .**

Q115 Written

The 3rd term is 30, and the 10th term is 2. Find a_n .

Classified Practice

Arithmetic Sequences

Q116 **Written**

The 10th term is 49, and the 23rd term is 10. Find a_n .

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Classified Practice

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(2n + 1, -1)$

B $(n + 3, 2)$

C $(a_n, a) = (2n - 1, 1)$

D $(2n - 1, 5)$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(13 - 2n, 15)$

B $(15 + 2n, 5)$

C $(a_n, a) = (15 - 2n, 13)$

D $(2n - 15, -13)$

Classified Practice

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(5n + 4, -1)$

B $(a_n, a) = (5n - 4, 1)$

C $(4n - 5, 2)$

D $(5n - 4, 31)$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n, a) = (3n - 10, -7)$

B $(3n - 4, -1)$

C $(2n - 7, -5)$

D $(10 - 3n, 7)$

Classified Practice

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Choose the correct option.

- A $\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$
B $\left(a_n = \frac{n+2}{2}, -\frac{5}{2}\right)$
C $\left(a_n = \frac{n-2}{2}, \frac{1}{2}\right)$
D $\left(a_n = \frac{2n-1}{3}, -\frac{1}{3}\right)$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Choose the correct option.

A $(a_n = 2n + 4, 2)$

B $(a_n = 2n + 2, 4)$

C $(a_n = n + 4, 5)$

D $(a_n = 2n - 2, 8)$

Classified Practice

Arithmetic Sequences

Q124 **Written**

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = -\frac{3(n+1)}{2}, -3\right)$

B $\left(a_n = -\frac{3(n-1)}{2}, -6\right)$

C $\left(a_n = \frac{3(n+1)}{2}, 3\right)$

D $\left(a_n = -\frac{n+1}{2}, -3\right)$

Classified Practice

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Choose the correct option.

A $\left(a_n = \frac{2n+4}{3}, 2\right)$

B $\left(a_n = \frac{2n-6}{3}, -\frac{8}{3}\right)$

C $\left(a_n = \frac{n+6}{3}, \frac{8}{3}\right)$

D $\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Classified Practice

Arithmetic Sequences

Q128 **Written**

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Classified Practice

Arithmetic Sequences

Q130 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Q131 MCQ**For $a = 3, d = 4$, which term of the arithmetic sequence is 59?:****Choose the correct option.****A $n = 14$** **B $n = 16$** **C $n = 15$** **D $n = 13$**

Classified Practice

Arithmetic Sequences

Q132 MCQ

For $a = -5$, $d = 6$, which term of the arithmetic sequence is 55?:

Choose the correct option.

A $n = 10$

B $n = 12$

C $n = 9$

D $n = 11$

Q133 MCQ

For $a = 20$, $d = -3$, which term of the arithmetic sequence is -16 ?:

Choose the correct option.

A $n = 13$

B $n = 12$

C $n = 14$

D $n = 11$

Classified Practice

Arithmetic Sequences

Q134 MCQ

For $a = 1, d = 5$, which term of the arithmetic sequence is 86?:

Choose the correct option.

A $n = 17$

B $n = 18$

C $n = 19$

D $n = 16$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Classified Practice

Arithmetic Sequences

Q136 **Written**

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Q137 MCQ**For $a = 2, d = 7$, find the first term greater than 100:****Choose the correct option.****A $n = 16$** **B $n = 15$** **C $n = 17$** **D $n = 14$**

Classified Practice

Arithmetic Sequences

Q138 MCQ

For $a = -9$, $d = 4$, find the first term greater than 50:

Choose the correct option.

A $n = 15$

B $n = 17$

C $n = 14$

D $n = 16$

Q139 MCQ**For $a = 6$, $d = 3$, find the first term greater than 200:****Choose the correct option.**

A $n = 65$

B $n = 67$

C $n = 64$

D $n = 66$

Classified Practice

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Choose the correct option.

A $n = 34$

B $n = 35$

C $n = 36$

D $n = 33$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Classified Practice

Arithmetic Sequences

Q142 **Written**

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Q143 MCQ**For $a = 10$, $d = -4$, find the first term smaller than -100 :****Choose the correct option.**

A $n = 28$

B $n = 29$

C $n = 30$

D $n = 27$

Classified Practice

Arithmetic Sequences

Q144 MCQ

For $a = 25$, $d = -6$, find the first term smaller than -250 :

Choose the correct option.

A $n = 46$

B $n = 47$

C $n = 48$

D $n = 45$

Q145 MCQ**For $a = -3, d = -5$, find the first term smaller than -200 :****Choose the correct option.**

A $n = 40$

B $n = 42$

C $n = 41$

D $n = 39$

Classified Practice

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Choose the correct option.

A $n = 17$

B $n = 16$

C $n = 18$

D $n = 15$

Q147 MCQ

For the expression For $a = 5, d = 4$, does the sequence contain the term 42?, the required value is:

Choose the correct option.

- A $n = 9$
- B No term
- C $n = 10$
- D $n = 11$

Classified Practice

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Q149 Written

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q150 **Written**

When x^2 , x , -8 form an arithmetic sequence in this order, find x .

Q151 Written

When $-3, x, x^2$ form an arithmetic sequence in this order, find x .

Classified Practice

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Classified Practice

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Classified Practice

Arithmetic Sequences

Q155 Written

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.

Q156 Written

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.

Classified Practice

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.**

Q158 MCQ**A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Choose the correct option.**

A 105

B 126

C 132

D 168

Classified Practice

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Choose the correct option.

A 64 B 72 C 81 D 144

Q160 MCQ**Find the sum of 10 terms whose first term is 15 and last term is 3:****Choose the correct option.**

A 45

B 90

C 150

D 180

Classified Practice

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Choose the correct option.

A 168

B 196

C 224

D 392

Q162 MCQ**Find the sum of 6 terms with first term -12 and last term 8 :****Choose the correct option.**A -120 B -12 C 12 D -2

Classified Practice

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Choose the correct option.

A 12

B 16

C 20

D 24

Q164 Written

Find (S_{13}) when the first term is (20), the common difference is (-5) , and the number of terms is (13).

Classified Practice

Arithmetic Sequences

Q165 **Written**

Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.

Q166 Written

Find (S_{15}) when the first term is (10), the common difference is (-3) , and there are (15) terms.

Classified Practice

Arithmetic Sequences

Q167 **Written**

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Q168 Written

Find (S_{18}) when the first term is (3), the common difference is (2), and there are (18) terms.

Classified Practice

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Choose the correct option.**

A 198

B 228

C 246

D 264

Q170 MCQ**For $a = 9, d = -2, n = 10$, find S_{10} :****Choose the correct option.****A** -90 **B** 0 **C** 9 **D** 90

Classified Practice

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Choose the correct option.**A -88 B 44 C 88 D 99

Q172 MCQ**For $a = 2, d = -3, n = 15$, find S_{15} :****Choose the correct option.****A** -315 **B** -285 **C** -270 **D** 285

Classified Practice

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Choose the correct option.**

A 11

B 44

C 88

D 96

Q174 MCQ**For $a = -3, d = -2, n = 7$, find S_7 :****Choose the correct option.****A** -42 **B** -54 **C** 54 **D** -63

Classified Practice

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

Classified Practice

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

Classified Practice

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Q180 MCQ**A sequence has $a = 5, d = 4, l = 41$. Find its sum:****Choose the correct option.****A** 205**B** 225**C** 230**D** 246

Classified Practice

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

Choose the correct option.

A 228

B 240

C 246

D 264

Q182 MCQ**A sequence has $a = 12, d = 2, l = 50$. Find its sum:****Choose the correct option.****A** 580**B** 600**C** 620**D** 640

Classified Practice

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10$, $d = 6$, $l = 50$. Find its sum:

Choose the correct option.

A 198

B 210

C 220

D 240

Q184 MCQ**A sequence has $a = 18, d = -3, l = -9$. Find its sum:****Choose the correct option.****A** -45 **B** 36 **C** 45 **D** 90

Classified Practice

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:**Choose the correct option.**

A 168

B 196

C 224

D 280

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Classified Practice

Arithmetic Sequences

Q187 **Written**

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Q188 MCQ**For $a = 3, d = 4$, find the sum from a_5 to a_{12} :****Choose the correct option.****A** 252**B** 264**C** 276**D** 288

Classified Practice

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :

Choose the correct option.

A -14

B 7

C 14

D 28

Q190 MCQ**For $a = 60, d = -3$, find the sum from a_8 to a_{16} :****Choose the correct option.**

A 216

B 234

C 243

D 270

Classified Practice

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :

Choose the correct option.

A 224

B 245

C 252

D 266

Q192 MCQ**For $a = 60, d = -3$, find the sum from a_9 to a_{17} :****Choose the correct option.****A 198****B 216****C 225****D 234**

Classified Practice

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Choose the correct option.

A 196

B 210

C 224

D 238

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Classified Practice

Arithmetic Sequences

Q195 **Written**

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Classified Practice

Arithmetic Sequences

Q197 MCQ

For $a = 4$, $d = 5$, find the sum from a_6 to a_{13} :**Choose the correct option.**

A 248

B 336

C 372

D 384

Q198 MCQ**For $a = -25$, $d = 6$, find the sum from a_5 to a_{11} :****Choose the correct option.****A** 105**B** 112**C** 119**D** 126

Classified Practice

Arithmetic Sequences

Q199 Written

Find (S_n) for a sequence with first term (1) and common difference (4).

Q200 Written**Find (S_n) for a sequence with first term (2) and common difference (-3) .**

Classified Practice

Arithmetic Sequences

Q201 **Written****Find (S_n) for a sequence with first term (0) and common difference (5).**

Q202 Written

Find (S_n) for a sequence with first term (3) and common difference (-4) .

Classified Practice

Arithmetic Sequences

Q203 Written

Find (S_n) for a sequence with first term (-4) and common difference (-2) .

Classified Practice

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Classified Practice

Arithmetic Sequences

Q205 **Written****Find the sum of the natural numbers from 1 to 100.**

Q206 Written**Find the sum of the natural numbers from 1 to 200.**

Classified Practice

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.**

Q208 Written**Find the sum of the natural numbers from 1 to 70.**

Classified Practice

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.

Q210 Written

Find the sum of the natural numbers from 1 to 400.

Classified Practice

Arithmetic Sequences

Q211 **Written****Find the sum of the natural numbers from 1 to 250.**

Q212 Written**Find the sum of the natural numbers from 1 to 81.**

Classified Practice

Arithmetic Sequences

Q213 **Written****Find the sum of the natural numbers from 1 to 301.**

Q214 Written**Find the sum of the natural numbers from 1 to 99.**

Classified Practice

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Choose the correct option.**

A 435

B 465

C 495

D 525

Q216 MCQ**Find the sum of the natural numbers from 1 to 45:****Choose the correct option.**

- A 990
- B 1035
- C 1080
- D 1125

Classified Practice

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Choose the correct option.**

A 1770

B 1830

C 1890

D 1950

Q218 MCQ**Find the sum of the natural numbers from 1 to 75:****Choose the correct option.****A** 2775**B** 2850**C** 2925**D** 3000

Classified Practice

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Choose the correct option.**

A 4005

B 4095

C 4185

D 4275

Q220 MCQ**Find the sum of the natural numbers from 1 to 125:****Choose the correct option.****A** 7750**B** 7875**C** 8000**D** 8125

Classified Practice

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Choose the correct option.**

A 16110

B 16290

C 16470

D 16650

Q222 MCQ**Find the sum of the natural numbers from 1 to 220:****Choose the correct option.****A** 24090**B** 24310**C** 24530**D** 24750

Classified Practice

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Choose the correct option.**

A 37675

B 37950

C 38225

D 38500

Q224 MCQ**Find the sum of the natural numbers from 1 to 350:****Choose the correct option.****A** 61075**B** 61425**C** 61775**D** 62125

Classified Practice

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.**

Q226 Written**Find the sum of the natural numbers from 201 to 300.**

Classified Practice

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.**

Q228 MCQ**A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Choose the correct option.****A** 133**B** 140**C** 147**D** 154

Classified Practice

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Choose the correct option.

A 351

B 360

C 369

D 378

Q230 MCQ**A sequence has first term -4, last term 20, and 9 terms. Select its sum:****Choose the correct option.****A** 63**B** 72**C** 81**D** 90

Classified Practice

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Choose the correct option.

A 188

B 196

C 204

D 212

Q232 MCQ**A sequence has first term 15, last term 3, and 10 terms. Select its sum:****Choose the correct option.****A** 80**B** 90**C** 100**D** 110

Classified Practice

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Choose the correct option.

A 507

B 520

C 533

D 546

Q234 MCQ**A sequence has first term 2, last term 57, and 12 terms. Select its sum:****Choose the correct option.****A** 342**B** 354**C** 366**D** 378

Classified Practice

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Choose the correct option.

- A 984
- B 1000
- C 1016
- D 1032

Q236 MCQ**A sequence has first term 101, last term 200, and 100 terms. Select its sum:****Choose the correct option.****A** 14950**B** 15050**C** 15150**D** 15250

Classified Practice

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Choose the correct option.

A 960

B 976

C 992

D 1008

Q238 Written**Find the sum of the multiples of 3 among the two-digit natural numbers.**

Classified Practice

Arithmetic Sequences

Q239 **Written****Find the sum of all two-digit natural numbers that are multiples of 5.**

Q240 Written**Find the sum of the multiples of 4 from 1 to 100.**

Classified Practice

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.**

Q242 MCQ**Select the sum of the multiples of 3 from 3 to 99:****Choose the correct option.****A** 1680**B** 1683**C** 1686**D** 1689

Classified Practice

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Choose the correct option.**

A 1296

B 1300

C 1304

D 1308

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.

Classified Practice

Arithmetic Sequences

Q245 **Written**

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Q246 Written**Find the sum of all two-digit odd natural numbers.**

Classified Practice

Arithmetic Sequences

Q247 **Written**

Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.

Classified Practice

Arithmetic Sequences

Q248 **Challenge****Find the sum of the numbers from 1 to 100 which are not divisible by 4.**

Classified Practice

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Q250 Written

Find n when $a = 3$, $d = 2$ and $S_n = 63$.

Classified Practice

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:**Choose the correct option.**

A 5

B 6

C 7

D 8

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Choose the correct option.

A 7

B 8

C 9

D 10

Classified Practice

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Choose the correct option.**

A 9

B 10

C 11

D 12

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Choose the correct option.**

A 11

B 12

C 13

D 14

Classified Practice

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:**Choose the correct option.**

A 14

B 15

C 16

D 17

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:**Choose the correct option.**

A 17

B 18

C 19

D 20

Classified Practice

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Choose the correct option.**

A 19

B 20

C 21

D 22

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Choose the correct option.**

A 23

B 24

C 25

D 26

Classified Practice

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:**Choose the correct option.**

A 27

B 28

C 29

D 30

Q260 MCQ**Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:****Choose the correct option.****A** 31**B** 32**C** 33**D** 34

Classified Practice

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:**Choose the correct option.**

A 35

B 36

C 37

D 38

Q262 MCQ**Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:****Choose the correct option.****A** 39**B** 40**C** 41**D** 42

Classified Practice

Arithmetic Sequences

Q263 **Written**

For first term 17 and common difference -2 , find how many terms have sum -63 .

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Classified Practice

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 80$:**Choose the correct option.**

A 3

B 4

C 5

D 6

Q266 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:****Choose the correct option.****A** 5**B** 6**C** 7**D** 8

Classified Practice

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10$, $d = 3$, find n when $S_n = 198$:**Choose the correct option.**

A 7

B 8

C 9

D 10

Q268 MCQ**Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:****Choose the correct option.****A** 9**B** 10**C** 11**D** 12

Classified Practice

Arithmetic Sequences

Q269 **Written**

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Classified Practice

Arithmetic Sequences

Q271 **Written****Find the general term with first term 4 and $S_6 = 114$.**

Q272 Written**Find the general term when $a_3 = 10$ and $S_{10} = 175$.**

Classified Practice

Arithmetic Sequences

Q273 **Written**

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q275 Written

For $a = 20$, $d = -6$, find n giving the maximum sum and the sum.

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Classified Practice

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.

Q278 Written**Find the general term where $S_6 = 3$ and $S_{10} = 65$.**

Classified Practice

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.**

Classified Practice

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Classified Practice

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Choose the correct option.

- A $\frac{11}{7}$
- B $\frac{10}{7}$
- C $\frac{11}{9}$
- D $\frac{9}{7}$

Quiz MCQ 2 Concept Quiz · MCQ

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Choose the correct option.

A $a_n = 4n + 2$

B $a_n = 4n - 2$

C $a_n = 5n - 5$

D $a_n = 2n + 4$

Classified Practice

Arithmetic Sequences

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Choose the correct option.

A 16

B 18

C 20

D 24

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Choose the correct option.

- A $\frac{1}{10}$
- B $\frac{1}{5}$
- C $\frac{5}{2}$
- D $\frac{1}{2}$

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 5 **Concept Quiz · Written**

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Classified Practice

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a) $d = 7, A = 22, B = 28; (b) d = 6, A = 19, B = 25; (c) d = -5, A = 14, B = -1; (d) d = -2, A = 13, B = 13; (e) d = 5, A = 14, B = 17; (f) d = -4, A = 13, B = 9, C = 1.$

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20; (b) a_n = 1 - 3n, a_{10} = -29; (c) a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}; (d) a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}.$

Q094 $a = 7, d = 3.$

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0, y = -2, z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	